SUNY Old Westbury 2026-2027 Undergraduate Catalog

Course Descriptions

Contract All Courses |

Biological Sciences

BS 1000 - Plants and Society

Credits: 4

A one semester lecture/laboratory course in general botany for non-science majors. The course covers the principles of plant anatomy and physiology, as well as the impact of plants and agriculture on society. Topics covered include the use and history of plants as food, medicines, and poisons.

BS 2100 - Biology for Non-science Majors

Credits: 4

A one-semester lecture/laboratory course in general biology for non-science majors. Surveys the major concepts and principles of biology, including cell structure and function, genetics, ecology, diversity and evolution. Topics related to the human experience are also discussed.

Prerequisites

EC II

When Offered:

Every semester.

BS 2200 - Food Safety: From Farm to Fork

Credits: 4

This is a 4-credit, 2000-level lecture course designed for biology majors and non-majors (including Public Health). Students learn core biological foundations needed to understand foodborne illness and chemical hazards, including: (i) a brief, accessible introduction to prokaryotic and eukaryotic cells; (ii) basic human physiology relevant to digestion, immunity, and inflammation; and (iii) how microbes, allergens, and selected contaminants affect human health across the food chain. The course emphasizes public health reasoning, outbreak thinking, risk communication, and evidence-based decision making. Class meetings use case studies, guided data interpretation, and short applied activities.

Prerequisites

None

BS 2300 - Anatomy and Physiology I Lecture

Credits: 3

This is the first semester of a two-semester sequence in which human anatomy and physiology are studied using a body systems approach with emphasis on the interrelationships between form and function at the gross and microscopic levels of organization. Demonstrating proficiency in this course is commonly required for entry into many Health Professions Programs such as Physician's Assistant, Physical Therapy, Occupational Therapy, and Nursing just to name a few. As such the course will be taught at a high level commensurate with ensuring such proficiency. Topics include; basic anatomical and directional terminology, fundamental concepts and principles of cell biology, histology and in depth study of the integumentary, skeletal, muscular and nervous systems.

Prerequisites

ECII and MA 1020

When Offered:

Every Semester

BS 2301 - Anatomy and Physiology I Laboratory

Credits: 1

This course is the laboratory component for BS 2300 - Anatomy and Physiology I Lecture

. Demonstrating proficiency in this course is commonly required for entry into many Health Professions Programs such as Physician's Assistant, Physical Therapy, Occupational Therapy, and Nursing just to name a few. As such the course will be taught at a high level commensurate with ensuring such proficiency. It covers the study of anatomical and directional terminology, fundamental concepts and principles of cell biology, histology and in depth study of the integumentary, skeletal, muscular and nervous systems using an experimental approach complemented with the use of preserved mammalian organisms as well as models.

Prerequisites

ECII and MA 1020

When Offered:

Every Semester

BS 2310 - Anatomy and Physiology II Lecture

Credits: 3

This is the second semester of a two-semester sequence in which human anatomy and physiology are studied using a body systems approach with emphasis on the interrelationships between form and function at the gross and microscopic levels of organization. Demonstrating proficiency in this course is commonly required for entry into many Health Professions Programs such as Physician's Assistant, Physical Therapy, Occupational Therapy, and Nursing just to name a few. As such the course will be taught at a high level commensurate with ensuring such proficiency. Topics include; in depth study of the endocrine system, digestive system, metabolism, nutrition, respiratory system, cardiovascular system, lymphatic system and immunity, urinary system, fluid/electrolyte and acid/base balance, and the male and female reproductive systems.

Prerequisites

BS 2300 and BS 2301 with a C or better

BS 2311 - Anatomy and Physiology II Laboratory

Credits: 1

This course is the laboratory component for BS 2310 - Anatomy and Physiology II Lecture

. Demonstrating proficiency in this course is commonly required for entry into many Health Professions Programs such as Physician's Assistant, Physical Therapy, Occupational Therapy, and Nursing just to name a few. As such the course will be taught at a high level commensurate with ensuring such proficiency. It covers the in depth study of the endocrine system, digestive system, metabolism, nutrition, respiratory system, cardiovascular system, lymphatic system and immunity, urinary system, fluid/electrolyte and acid/base balance, and the male and female reproductive systems. These topics will be covered using an experimental approach complemented with the use of preserved mammalian organisms as well as models.

Prerequisites

BS 2300 and BS 2301 with a C or better

When Offered:

Every Semester

BS 2400 - Basic Biosciences I

Credits: 3

First semester of freshman biology courses covering content areas including molecular and cellular biology, genetics, genomics and evolution.

Prerequisites

ECI**Corequisite** BS2401 if taken for the first time**Prerequisite/Corequisite**

BS2401

When Offered:

Every Semester

BS 2401 - Basic Biosciences I Lab

Credits: 1

First semester of freshman biology laboratory courses covering content areas including molecular and cellular biology, genetics, genomics and evolution.

Prerequisites

ECI**Corequisite** BS2400 if taken for the first time**Prerequisite/Corequisite**

BS2400

When Offered:

Every Semester

BS 2410 - Basic Biosciences II

Credits: 3

Second semester of freshman biology courses covering content areas including cell metabolism, comparative animal and plant anatomy and physiology and ecology.

Prerequisites

BS 2400 and BS 2401 with a C or better; MA 1020 or math proficiency**Corequisite** BS 2411 if taken for the first time**Prerequisite/Corequisite**

BS 2411

When Offered:

Every Semester

BS 2411 - Basic Biosciences II Lab

Credits: 1

Second semester of freshman biology laboratory courses covering content areas including cell metabolism, comparative animal and plant anatomy and physiology and ecology.

Prerequisites

BS 2400 and BS 2401 with a C or better; MA 1020 or math proficiency**Corequisite** BS 2410 if taken for the first time**Prerequisite/Corequisite**

BS 2410

When Offered:

Every Semester

BS 2490 - Environmental Science

Credits: 4

The focus of this lecture course is on the interactions of humans with their environment. Human impact on the world's ecosystems is emphasized. The theme stressed throughout this course is that every system involving life is dependent upon maintaining balances between nutrient gains and losses, between energy inflows and outflows, and between births and deaths within populations. Against this background, various aspects of the human system, such as our use of soils, our disposal of wastes, our use of resources, land, and energy and our population, are shown to be out of balance. The consequences of these imbalances are discussed in relation to our political and economic system.

Prerequisites

ECII

When Offered:

Every Fall and Spring

BS 3300 - Biotechnology Workshop

Credits: 4

Designed to develop laboratory skills in the basic techniques used to study molecular genetics. Both prokaryotic and eukaryotic organisms are used as models for the experiments. Some of the techniques introduced in the course include DNA isolation, agarose gel electrophoresis, restriction enzymes, gene transfer, gene cloning, southern blot, PCR, RT-PCR, and gene sequencing. In the latter part of the course, students will be introduced to the Human Genome Project and to some of the techniques used in DNA sequencing and analysis.

Prerequisites

EMS

When Offered:

Occasionally.

BS 3400 - Vertebrate Physiology

Credits: 4

A regulatory systems approach to the principles of animal physiology, developed through consideration of functioning of cells, tissues, and organ systems with an emphasis on humans. (With 3-hour laboratory)

Prerequisites

BS 2400, BS 2401, BS 2410, BS 2411 with a C or better; CP 2120 and CP 2121 with a C or better; ECII

When Offered:

Every semester

BS 3450 - Plant Biology

Credits: 4

The study of plants as biological organisms. Topics include classification and evolution of the plant kingdom: plant anatomy, physiology and reproduction, with emphasis on the flowering plants. Consideration given to experimental methods used in studying plants and the importance of plants and plant studies. Students are expected to complete an experimental project or observational study. Lecture, discussion, laboratory and field trips, or audio tutorial integration of these are presented.

Prerequisites

BS 2400, BS 2401, BS 2410, BS 2411 with a C or better, ECII

When Offered:

Every Fall

BS 3500 - Invertebrate Zoology

Credits: 4

This lecture/laboratory course introduces the student to the diversity of form and function among the invertebrates using a phylogenetic approach. It emphasizes their importance in ecology, diseases, food sources and as biomedical models. Laboratory sessions include dissection of representative animals and encourage individual research projects.

Prerequisites

BS 2400 - BS 2410 completed with grades of “C” or better

BS 3510 - Parasitology

Credits: 4

Lecture/laboratory course. Major parasites of medical and economic interest are examined, including the causal agents of malaria, trypanosomiasis schistosomiasis, and filariasis. The ecology, anatomy, pathology, and public health importance of these parasites are emphasized. Laboratories include both live and prepared material.

Prerequisites

BS 2400 - BS 2410 with grades of "C" or better; EC II

BS 3520 - Comparative Anatomy

Credits: 4

Lecture/laboratory course designed for biology majors and premedical students. Emphasis is on the basic anatomical patterns of vertebrate structure. Morphogenesis and structure function relationships are discussed in terms of adaptive significance. Laboratory includes dissections of aquatic and terrestrial vertebrates.

Prerequisites

BS 2400, BS 2401, BS 2410, BS 2411 with a C or better; ECII

When Offered:

Every Fall

BS 3530 - Comparative Genomics

Credits: 4

Lecture/laboratory course. Explore the interdisciplinary field of genome science. Study the genomic similarities and differences between organisms, how the genome is constructed and operates, and how genomes evolved. Understand the advances in genomic technologies and their applications in health care. Lab sections will explore the latest sequencing technologies and tools to analyze genomic data.

Prerequisites

BS 2400, BS 2401, BS 2410, BS2411 with a C or better, ECII

When Offered:

Every Fall**Note:**

This course is the combined comparative genomics lecture and lab, and is part of the Biological Sciences BA/BS and Bioinformatics and Computational Biology BS. These degrees will be updated to reflect the combined lecture/lab.

BS 3531 - Comparative Genomics Lab

Credits: 4

Lecture/laboratory course. Explore the interdisciplinary field of genome science. Study the genomic similarities and differences between organisms, how the genome is constructed and operates, and how genomes evolved. Understand the advances in genomic technologies and their applications in health care. Lab sections will explore the latest sequencing technologies and tools to analyze genomic data.

Prerequisites

BS 2400/BS 2401 andBS 2410/BS 2411 with grade of "C" or better.

When Offered:

Every other semester.

BS 3810 - Biological Aspects of Aging

Credits: 4

Lecture/workshop course designed for science students. Examines the aging process on the molecular, cellular, and organismic levels. Emphasizes alterations during the aging process in cells whose life span is identical to that of the entire organism. Examination of various pathological disorders in relationship to aging is also emphasized. Students are required to read original articles and be prepared to discuss current topics in workshop settings. Completion of BS3400 is recommended prior to enrolling in this course.

Prerequisites

BS2400, BS 2401, BS 2410, BS 2411 with a C or better, ECII

When Offered:

Every Spring.

BS 3910 - Intro to Bioinformatics

Credits: 4

Lecture/laboratory course. Explore the application of computer science tools to address biological questions through lectures and labs. Survey the genomic and biomolecular databases and web bioinformatics resources. Understand the algorithms and learn basic programming languages to analyze DNA, RNA, and proteins. Apply bioinformatics tools to analyze real-world problems and data.

Prerequisites

BS 2400, BS 2401, BS 2410, BS 2411 with a C or better; ECII

When Offered:

Every Spring

BS 4400 - Cell Biology

Credits: 4

Lecture/discussion/laboratory course whose content is directed to the study of the molecular basis of cell activities, cell ultra structure, and its integration with cellular function. Lectures survey the cell in its environment, energy transformation, mobility, replication of genetic material, growth, nucleocytoplasmic interactions. The laboratory explores techniques used in the study of the cell.

Prerequisites

BS 2400, BS 2401, BS 2410, BS 2411 with a C or better; ECII

When Offered:

Every Fall and Spring

BS 4410 - Histology

Credits: 4

A laboratory-oriented course dealing with the microscopic and ultramicroscopic anatomy of mammalian tissues and organs with emphasis on relating structure to function.

Prerequisites

BS2400, BS2401, BS2410, BS2411 with a C or better; ECII

When Offered:

Every Fall

BS 4420 - Microbiology

Credits: 4

Lecture/laboratory course concerned with structure-function relationships in various types of microorganisms, with particular emphasis on bacteria. Nutrition, biosynthesis, energetics, growth, microbial interactions and symbiotic relationships, host-parasite relationships, and microbial genetics are among the topics discussed.

Prerequisites

BS 2400, BS2401, BS2410, and BS2411 with a grade of "C" or better; EC Il

BS 4430 - Developmental Biology

Credits: 4

A lecture/laboratory course covering fundamental principles of developmental biology. Topics included: gametogenesis, fertilization, cleavage, gastrulation, growth and differentiation. This course attempts to integrate the descriptive, experimental and genetic approaches to the study of developmental biology. Completion of BS 3520 is recommended.

Prerequisites

BS 2400, BS 2401, BS 2410, BS 2411 with a C or better; ECII

When Offered:

Every Spring

BS 4440 - Evolution

Credits: 4

Lecture/discussion course presenting the history of life on earth from its inception to the emergence and early history of Homo sapiens. Through readings, lectures, demonstrations, laboratories, discussions, and museum trips, students examine modern evolutionary theory and concepts, including abiogenesis, adaptation, population genetics, race, speciation, and progressive evolution. Completion of BS4460 is recommended.

Prerequisites

BS 2400, BS 2401, BS2410, and BS 2411 with a C or better; ECII

When Offered:

Every Fall and Spring

BS 4460 - Genetics

Credits: 4

Lecture/laboratory course presenting the fundamental principles of genetics, the molecular biology of the gene, and heredity in humans. Through lectures, readings, laboratories, and discussions, students examine the experimental evidence leading to currently accepted concepts, and critically analyze the implications of various findings in human genetics.

Prerequisites

BS 2400, BS 2401, BS 2410, and BS 2411 with a C or better; ECII

When Offered:

Every Fall and Spring

BS 4461 - Immunology

Credits: 4

Lecture/laboratory course. This course will introduce students to the essential concepts of how the immune system works, what are the molecular and cellular mechanisms and pathways involved in the defense of an organism from infectious diseases or cancer, and it will describe in detail the cells and molecules of the mammalian immune system. The laboratory component is a research-based course where students participate in the development of new knowledge by generating a novel research question and performing their own experimental design. Completion of CP 2120 and BS 4400 or BS 4460 is recommended.

Prerequisites

BS 2400, BS 2401, BS 2410, and BS 2411 with a C or better; ECII

When Offered:

Occasionally

BS 4470 - Ecology

Credits: 4

A general ecology course focused on the study of ecosystems. It fulfills the population content area course requirement for the BA and BS degrees in Biology. The course includes a lecture (3 hours/week) and a laboratory component (3 hours/week). Topics include ecosystem structure and function, population dynamics, community structure, food webs, energy flow, nutrient cycles, biogeography, associations, succession, adaptation and niche analysis, and interspecific interactions. The course is taught in a blended format with a significant online course component. The laboratory component includes extensive field work. Students working in groups must design and implement a research project involving data collection from a campus ecosystem. The project culminates in a group oral presentation. Students are also encouraged to present their results at a local or regional conference and/or submit their results for publication to a peer-reviewed undergraduate research journal

Prerequisites

Bs 2400, BS 2401, BS 2410, and BS 2411 with a grade of C or better; ECII

When Offered:

Every Fall

BS 4474 - Microbial Ecology

Credits: 4

This course is designed to introduce students to the study of microbial ecosystems its structure and function using an applied learning research approach modeling the scientific research process. Students should be aware this course requires a significant amount of independent work. In the lecture component students will be introduced to ecological theory to support the laboratory experimentation. As part of their investigation the students will learn about ecosystem structure, i.e. biotic and abiotic components, and function, i.e. symbiotic interactions and food webs. As part of the biotic

component of the ecosystem, the students will focus on the study of a microbiome using both traditional microbiology and next generation sequencing methodology, i.e. microscopy, isolation using generalized and selective media, and molecular techniques: DNA isolation from environmental samples, PCR amplification using specific primers, i.e. 16S rDNA, and sequence analysis. This course fulfills the population core requirement for the BA and BS in Biology, and can also be used as one of the upper division biology course requirement in the BS in Bioinformatics and Computational Biology. This course is recommended for students who plan to pursue a career in basic science or biomedical research, and the health profession. Prerequisite: BS2400/BS2401-2410/BS2411 with a grade of C or better.

Prerequisites

BS 2400, BS 2401, BS 2410, and BS 2411 with a C or better; ECII

When Offered:

Every Spring

BS 4480 - Animal Behavior

Credits: 4

A lecture/laboratory course introducing the field of animal behavior. Lectures, discussions and films explore topics including the cultural basis of behavior, the organization and physiology of the nervous system and its role in controlling behavior, the endocrine basis of behavior, genetic basis of behavior and sociobiology.

Prerequisites

BS 2400, BS 2401, BS 2410, and BS 2411 with a C or better; ECII

When Offered:

Occasionally

BS 4491 - Human Ecology

Credits: 4

Students will examine the adaptation of human populations to their environment. Topics include: biological and cultural evolution, human diversity, impact of limiting resources, the ecology of disease, and urban ecology and planning.

Prerequisites

BS 2400/BS 2401 and BS 2410/BS 2411 with grades of "C" or better

When Offered:

This course is offered once a year.

BS 4500 - Neurobiology

Credits: 4

Lecture course designed to familiarize students with cellular mechanisms within the nervous system and how they are used to promote intercellular communication. Connecting molecular signaling to higher-order neurobiological function will be a large focus of the course. Includes concepts as they relate to neurobiology such as synaptic transmission, neuroplasticity, neuronal development, sensory systems, neuropharmacology, drugs of abuse, and neurodegenerative/neurodevelopmental disorders. Students will prepare a research project utilizing primary scientific literature and will be expected to convey this information via writing. Completion of BS 4400 is recommended.

Prerequisites

BS 2400, BS 2401, BS 2410, BS 2411 with a C or better; ECII

When Offered:

Fall, every other year.

BS 4550 - Cancer Cell Biology

Credits: 4

Lecture/Discussion. The study of cancer disease requires knowledge of several essential biological processes that govern the basic of the living system. This course will explore the cellular and molecular mechanisms underlying cancer development with the aim of understanding how changes in the normal cell growth and division processes lead to the formation of tumors. Case studies will be used to explore the roles of specific mutations in cancer develop- ment and the pathways they affect, as well as the classes of genes targeted during cancer development. This upper level course builds on concepts introduced in courses in Cell Biology and Genetics and will emphasize experimental methods and design as well as critical data analysis. Students will discuss papers from the primary literature each week individually or in groups. Completion of BS 4400 or BS 4460 is recommended.

Prerequisites

BS 2400, BS 2401, BS 2410, and BS 2411 with a C or better; ECII

When Offered:

Occasionally

BS 4560 - Molecular Biology

Credits: 4

Lecture/Laboratory course to introduce students to both the theoretical and practical aspects of molecular biology. An understanding of molecular biology necessitates a synthesis of information from genetics, biochemistry, cell biology, physics and chemistry. The course will include microbial studies, cellular components, genes, regulation etc. The lab component will utilize techniques of genetic engineering.

Prerequisites

BS 2400, BS 2401, BS 2410, and BS 2411 with a C or better; CP 2120 with a C or better; ECII

When Offered:

Every Spring

BS 4565 - Pharmacology

Credits: 4

A lecture/readings course dealing with the principles of drug interactions in animal cells and tissues. The molecular level of pharmacological actions and their subsequent impact on behavioral and physiological functions are studied. Students are required to read primary scientific literature and are expected to be able to convey the information contained within through writing. This course is intended for students who plan for a career in a variety of fields, ranging from allied health professions through basic science research.

Prerequisites

BS 2400, BS 2401, BS 2410, BS2411 with a C or better, ECII

When Offered:

Other**Note:**

This course has run three times so far, and already is in Banner and DegreeWorks. It is not present in the course catalog and this is to rectify that discrepancy.

BS 4680 - Environmental Health

Credits: 4

This course takes an integrated approach to the development of a basic understanding of the impact of the environment on health. It examines chronic and infectious diseases in relation to modern life styles. A new section examines the AIDS epidemic in relation to the environment. Other topics that are covered include: air pollution in the general industrial environment; water pollution; the effects of pesticides and food additives; synthetic organic chemicals such as PCB's and dioxins as environmental estrogens; health effects of radiation and safety of nuclear energy. Satisfies elective requirement for B.A. in Biology.

Prerequisites

EC II

When Offered:

On demand.

BS 5400 - Senior Seminar

Credits: 4

Senior Seminar in Biology is designed to promote an in-depth consideration of special topics in current biological research. The student will, under the advisement of a faculty mentor, develop a topic as a library study and present the results both orally and as a written paper. The process of writing this paper will consist of a series of tasks that must be completed in order. The course will also include presentations by external speakers, and students will be asked to read papers of these speakers, draft questions to ask the speakers, and to write short commentary in regard to each speaker. This course is intended to be taken in the final semester before graduation. Student eligibility will be determined by faculty review.

Prerequisites

None, requires permission of instructor or Chair to register

When Offered:

Every Semester**Note:**

Senior Seminar has already run in Fall 2024, and is in Banner and DegreeWorks. This is to update the active course catalog.

This course is intended to be taken in the final semester before graduation. Students are individually screened by faculty before registering for the course to determine eligibility.

BS 5410 - Seminar I in Biology: Reading in the Discipline

Credits: 2

Designed to increase ability to read in the discipline. Students will survey scientific literature, learn how to research topics, prepare abstracts, etc. Students will select a mentor and topic for Seminar II. Must be taken by students two semesters prior to graduation.

Prerequisites

85-90 credits and permission of instructor

When Offered:

Every Fall and Spring.

BS 5420 - Seminar II in Biology: Writing in the Discipline

Credits: 2

Designed to promote an in-depth consideration of special topics in current biological research. The course will include presentations by both external speakers and students. The student will, under the advisement of a faculty mentor, develop a topic either as a library or laboratory study and present the results both orally and as a written paper.

Prerequisites

BS 5410

When Offered:

Every Fall and Spring.

BS 5590 - Advanced Research

Credits: 4

Research project carried out under the supervision of a faculty member. Each project includes a survey of the appropriate original literature and investigation of a selected research topic by experimental or theoretical methods. Open to students of upper-division standing, preferably seniors, on acceptance as a research student by a faculty member. Students may take this course twice for credit but may receive only 4 credits from this course toward the major. Grading will be CR/NC.

Prerequisites

BS 2400 - BS 2410 with grades of "C" or better; permission of instructor

When Offered:

Every semester.

BS 5591 - Environmental Research

Credits: 4

Research project in environmental science carried out under supervision of a faculty member. Open to upper-division students, preferably seniors on acceptance as a research student by a faculty member.

Prerequisites

BS 2400 - BS 2410 with grades of "C" or better; 3 upper-division Biology courses; permission of instructor

When Offered:

Every semester.

BS 6750 - Biological Pharmacology

Credits: 3

A lecture/readings course dealing with the priciples of drug interactions in animal cells and tissues. The molecular and behavioral levels of pharmacological actions with impact on physiological functions are studied. Substance abuse issues will be highlighted.

Prerequisites

Successful completion of the core science courses with a grade of B or better. (BS 4400 or BS4460 and BS 3400 or BS3520.

BS 9990 - Independent Study

Credits: 4

Independent study is to be carried out under the supervision of a faculty member. It permits students to investigate areas that are not part of existing offerings.

Prerequisites

BS 2400 - BS 2410 with grades of "C" or better; permission of instructor

When Offered:

Open to students of upper-division standing only on acceptance by a faculty member, and approval by department chair.

Contract All Courses |